

25 Most Important Quality Metrics (KPI's)

©Manickavasagam Natarajan



Customer Quality KPI's

Sr. No.	KPI	Calculation Method	Ideal Range
1	Customer PPM / External PPM	$\frac{\text{No of Defective Parts}}{\text{No of Parts supplied}} \times 1 \text{ million}$	Minimum (Lower the Better)
2	Customer Rejection (%)	$\frac{\text{No of Defective Parts}}{\text{No of Parts supplied}} \times 100$	Minimum (Lower the Better)
3	No of Customer Complaints per Period	Total Complaints during the defined time	Minimum (Lower the Better)
4	Average Time to solve (Hrs. / Days)	Time between complaint receipt and complaint closure - Correction, CAPA.	Minimum (Lower the Better)
5	Average Time to Respond (Hrs. / Days)	Time between complaint receipt and response provided to customer	Minimum (Lower the Better)
6	Customer Satisfaction / Net Promoter Score (NPS)	$\text{NPS} = \% \text{ Promoter} - \% \text{ Detractors}$	Maximum (Higher the Better)

Follow @Manickavasagam Natarajan

Supplier Quality KPI's

©Manickavasagam Natarajan

Sr. No.	KPI	Calculation Method	Ideal Range
7	Supplier PPM	$\frac{\text{No of Defective Parts}}{\text{No of Parts Received}} \times 1 \text{ million}$	Minimum (Lower the Better)
8	Supplier Rejection / Defect Rate (%)	$\frac{\text{No of Defective Parts}}{\text{No of Parts Received}} \times 100$	Minimum (Lower the Better)
9	OTD - On time Delivery	$\frac{\text{Total deliveries on time}}{\text{Total deliveries made}} \times 100$	Maximum (Higher the Better)
10	OTIF - On Time in Full	$\frac{\text{Total deliveries on time in full}}{\text{Total deliveries made}} \times 100$	Maximum (Higher the Better)
11	Average Time to Respond (Hrs. / Days)	Time between complaint / Query receipt and response provided	Minimum (Lower the Better)
12	Average Time to solve (Hrs. / Days)	Time between complaint receipt and closure - Correction, CAPA	Minimum (Lower the Better)

Incoming / Receiving Quality - IQC KPI's

Sr. No.	KPI	Calculation Method	Ideal Range
13	Average Time to Inspect (Hrs. / Days)	Time between Part receipt and Inspection Completion	Minimum (Lower the Better)
14	Average Time to raise rejection note / QN (Hrs. / Days)	Time between physical material rejection and Rejection Note / QN- Quality Notification Creation	Minimum (Lower the Better)

Follow @Manickavasagam Natarajan

General - Quality Planning, Management & Improvement

©Manickavasagam Natarajan

Sr. No.	KPI	Calculation Method	Ideal Range
15	Cost of Good Quality - COGQ	$\frac{(\text{Appraisal Cost} + \text{Prevention Cost})}{\text{Total Company Revenue}}$	Minimum (Lower the Better)
16	Cost of Poor Quality - COPQ	$\frac{(\text{Internal Failure Cost} + \text{External Failure Cost})}{\text{Total Company Revenue}}$	Minimum (Lower the Better)
17	Cost of Quality - COQ	$\frac{(\text{COPQ} + \text{COGQ})}{\text{Total Company Revenue}}$	Minimum (Lower the Better)
18	No of Training completed	Total Trainings completed in the Targeted Time	Maximum (Higher the Better)
19	No of Non-Conformance NC's per area (Internal / External Audit)	Total NC's observed in the External & Internal Audits (Certification Body's, Customers, Regulatory etc) in the Targeted Time	Minimum (Lower the Better)
20	Average Days for closing Audit observations	Time between Audit observation and Closure of Observations.	Minimum (Lower the Better)

Follow @Manickavasagam Natarajan

Inprocess Quality KPI's

©Manickavasagam Natarajan

Sr. No.	KPI	Calculation Method	Ideal Range
21	Internal PPM	$\frac{\text{No of Defective Parts}}{\text{No of Parts Produced}} \times 1 \text{ million}$	Minimum (Lower the Better)
22	Rejection / Defect Rate (%)	$\frac{\text{No of Defective Parts}}{\text{No of Parts Produced}} \times 100$	Minimum (Lower the Better)
23	Scrap Rate %	$\frac{\text{Total Scrap}}{\text{Total Product}} \times 100$	Minimum (Lower the Better)
24	Yield / Efficiency	$\text{Yeild} = (I)(G) + (I)(1-G)(R)$ I = planned production G = correct parts [%] R = reworked parts [%]	Maximum (Higher the Better)
25	Throughout	$\frac{\text{Correct Part}}{\text{Total Production Time}} \times 100$	Target

If you find it useful, please     

Follow **Manickavasagam Natarajan** for more exciting posts on
 Quality, Safety, Operational Excellence, Leadership, and
 Inspiration.   